

Optimal trajectory generation for wheeled mobile robot

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Abstract. The problem associated with trajectory generation for a car-like robot (CLR) moving from an initial configuration to any final configuration is studied in this paper. The aim of our research is to achieve energy-optimal solution to the problem of trajectory generation. Based on bang-bang control theory, the necessary condition for energy-optimal trajectory generation is studied. Based on steering angle velocity control model (SAVCM), the presented method can be used to deal with the trajectory generation problem of CLR whose steering angle is not instantaneously realizable effectively.

1 Introduction

Steering angle control model (SACM) is often used to solve the problem of trajectory generation for CLR with fixed rear driving wheels and front steering wheels, see [1][7][8-16]. To a CLR system, if we concern on the constant velocity motion on a plane, the control should be steering angle or velocity of steering angle only. From the existed literatures of this research area, it is found that most of the research results were based on the SACM but not the SAVCM. If we use SACM, the configuration of the mobile robot considered is a vector of (x, y, θ) , where (x, y) is the coordinate of a reference point describing the position of the vehicle and θ is the orientation with respect to the x-axis. In terms of this kind of dynamic model, the optimal path planning problems were addressed on shortest paths problem mostly. The first study on this problem was made by Dubins [13] who proved that the existent shortest paths are at most made of three pieces which are straight line segments or arcs of circle with minimal radius. Following this result, Cockayne and Hall [14] computed different kinds of paths and the set of reachable positions; Reeds and Shepp extended Dubin's result to the system which can moves forward and backward with constant velocity [15]; Boissonnat *et al* gave a proof to Reeds and Shepp's result; Soueres and Laumond analyzed the results theoretically [7]. Shortest distance paths from prescribed initial configuration to final configuration were generated by Alexander *et al* 1998 [1] using difference perspective of distance which is defined from any picked point on the mobile robot to a prescribed point.

If the steering angle is not instantaneously realizable, i.e. the time spent to change the steering angle from an initial configuration to a prescribed one is not short (such as 0.5 second, in which the vehicle may move several meters), what will happen? The answer is that the trajectory will never be realized by any trajectory-following controller. The reason is the using of steering angle to control the mobile robot. The

difficulty will be avoided if we employ the steering angle velocity control or we take no concern with optimal performance. For the systems of omnidirectional and skid-steer platforms, this problem is not as significant as that for a CLR, see Killoughh's work [3]. To the literatures mentioned above, this problem cannot be avoided unless the motion velocity is very small with respect to velocity of steering angle. In the case when a CLR moves with a high speed, the mv/sv (moving velocity/steering angle velocity) ratio has strong and significant effect on the motion dynamics. The steering angle control is not reasonable for a CLR. Therefore, we employ SAVCM to study the problem of trajectory generation of CLR.

The energy-optimal trajectory generation problem for a CLR is equal to least-swing control problem. Pontrygin's maximum principle (PMP) has been used to deal with optimal problem by most researchers. For examples, Alexander *et al* [1] studied the necessary condition for two-point boundary-value problem of minimal distance paths by using PMP as well as Reister and Pin [2] proved that the solution to the problem of time-optimal trajectory generation were bang-bang for a skid-steer mobile robot based on PMP. More literatures about this research area can be found, such as [7][12]-[14].

2 Problem Statement

The CLR considered in this paper is equipped with two steering wheels and two driving wheels which satisfies the following assumptions: The chassis and the rods linking the wheels to the chassis are both perfectly rigid; the wheels are nondeformable and have the same radii; each wheel obeys the customary pure rolling and nonslipping assumptions according to which the velocity of the contact point of each wheel with the ground is equal to zero. Moreover, it is assumed that the CLR moves on a horizontal plane. The schematic structure of a general CLR is illustrated in Figure 1. The variables and structural relationship can be seen from Figure 2.

We can use a simplified two-wheel bicycle like structure to illustrate the kinematic-dynamic constraint of the CLR, which is formed as (SAVCM),

$$\begin{aligned}\dot{x} &= v \cos \theta \\ \dot{y} &= v \sin \theta \\ \dot{\theta} &= v \tan \phi, -\phi_{\max} \leq \phi \leq \phi_{\max} \\ \dot{\phi} &= \alpha, \alpha_{\max} \leq \alpha \leq \alpha_{\max}\end{aligned}\tag{1}$$

The unique control variable is α which is the angular velocity of steering angle. The state vector is $(x, y, \theta, \phi)^T$ where (x, y) is coordinate at the reference point, θ is the orientation angle with respect to the x-axis, ϕ is the steering angle, v is the fixed velocity of the CLR. See Figure 2.

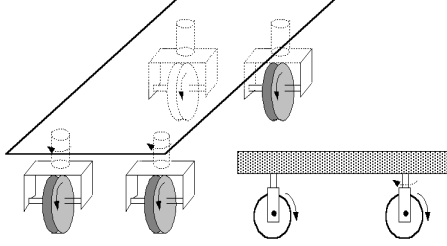


Fig.1: (Schematic structure of CLR)

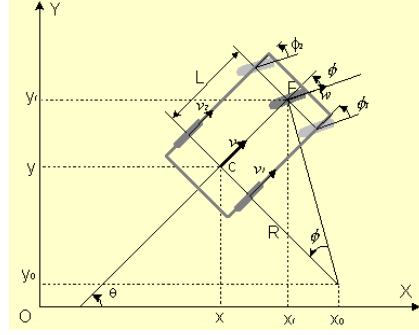


Fig.2: (The configuration of CLR)

Definition 1 The moving velocity divided by steering angle velocity, i.e. v/α , is denoted by mv/sv ratio.

Definition 2 The time in which the steering wheels change from an initial angle to an expected angle is called steering angle delay, is denoted by SAD.

Property 1 The bigger the mv/sv ratio the vehicle has, the larger the trajectory distance β is, where $\beta = \|\zeta_1 - \zeta_2\|_2$, $\zeta_1 = (x_{\zeta_1}(t), y_{\zeta_1}(t), \theta_{\zeta_1}(t))^T, t \in [0, t_f]$, is a trajectory generated with consideration of SAD (i.e. using the SAVCM), $\zeta_2 = (x_{\zeta_2}(t), y_{\zeta_2}(t), \theta_{\zeta_2}(t))^T, t \in [0, t_f]$, is a trajectory generated without consideration of SAD (i.e. using the SACM). $\|\cdot\|_2$ represents Euclidean norm.

This property can be illustrated by Simulation 1 readily. In Figure 3, the two curves represent the Cartesian trajectories generated by using SAVCM (1) and the following steering angle control model (SACM) respectively,

$$\begin{aligned}\dot{x} &= v \cos \theta \\ \dot{y} &= v \sin \theta \\ \dot{\theta} &= \beta\end{aligned}\tag{2}$$

where β is control variable, $(x, y, \theta, \phi)^T$ is state vector, x and y are the position coordinates at the reference point, θ is the orientation angle with respect to the x -axis, v is the fixed velocity of the CLR.

Because β has the description of $\beta = v \tan \phi$, so this model can be called SACM.

In the simulation, we compare the trajectory generated by SAVCM with the trajectory by SACM. Because the steering angle of CLR system can not be realized instantaneously, the trajectories generated by using the two models are quite different which is shown in Figure 3. It can be deduced that there exists an assumption that the steering angle is realizable instantaneously in then SACM. Therefore, if we use SACM to generate a trajectory, the SAD of the CLR is ignored. But if the mv/sv ratio is not very small, the ignoring of SAD is not reasonable. On the other hand, SAVCM can not only describes the kinematic performance but also the dynamic performance of steering angle. As for most practical CRL systems,

mv/sv ratio is big i.e. SAD cannot be ignored, so the SACM based trajectory generation method is invalid for CLR usually.

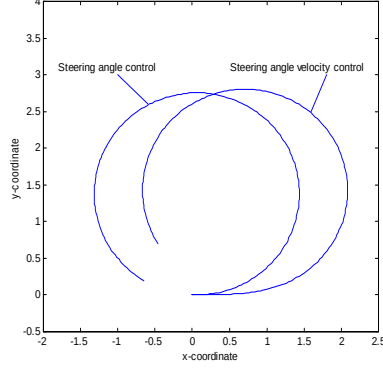


Fig.3: (Cures of position of trajectories)

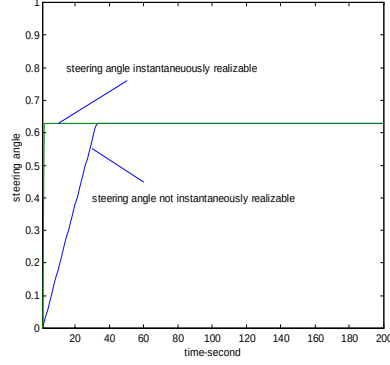


Fig.4: (Steering angles varying)

The optimal control problem investigated in this paper is formulated as the following:

Find the control function $\alpha(t)$ which minimizes the indicator function

$$J = \int_{t_0}^{t_f} |\alpha| dt \quad (3)$$

such that the solution, $x(t)$, $y(t)$, $\theta(t)$, $\phi(t)$, of differential equation (1), transfer from the initial value

$$x(t_0) = 0, y(t_0) = 0, \theta(t_0) = 0, \phi(t_0) = 0 \quad (4)$$

to the final value

$$x(t_f) = x_{t_f}, y(t_f) = y_{t_f}, \theta(t_f) = \theta_{t_f}, \phi(t_f) = 0 \quad (5)$$

where $t_0 = 0$ is the initial time and t_f is free final time.

It can be seen that our energy-optimal trajectory generation problem is different from previous description of problem of optimal trajectory generation.

3 Trajectory Generation with Bang-Bang Control

Alike most researchers did [7][12]-[14], we use PMP to deal with the optimal-control problem. In the following sections, the bang-bang control characteristic of the optimal-control problem (3)-(5) is introduced firstly. Then a 3-time switch control is used to generate the turning trajectory to replace the circle trajectory by using SAVCM so that the SAD can be considered.

3.1 General form of the optimal control

The Hamilton function for the optimal-control problem of (3)-(5) is

$$H[X(t), u(t), P(t), t] = P_1 v \cos \theta + P_2 v \sin \theta + P_3 v \tan \varphi + P_4 \alpha + |\alpha| \quad (6)$$

where $P = (P_1, P_2, P_3, P_4)^T$ is the costate.

Because PMP requires that

$$|\alpha^*(t)| + P_4^*(t)\alpha^*(t) \leq |\alpha(t)| + P_4^*(t)\alpha(t) \quad (7)$$

where α^* is an optimal control, P_4^* is the forth component of optimal costate [4], the following bang-bang control law corresponding to the above optimal-control problem can be obtained readily,

$$\alpha^*(t) = \begin{cases} \alpha_{\max} & \text{for } P_4^* < -1 \\ 0 & \text{for } -1 < P_4^* < 1 \\ -\alpha_{\max} & \text{for } P_4^* > 1 \end{cases} \quad (8)$$

$$\begin{aligned} & (-\alpha_{\max}, 0) \quad \text{for } P_4^* = 1 \\ & (0, \alpha_{\max}) \quad \text{for } P_4^* = -1 \end{aligned}$$

This gives us a guideline to generate an energy-optimal trajectory for our WMR system by using bang-bang control. In the next subsection we will study how to design a trajectory by the optimal control approach.

3.2 Bang-bang Trajectory Generation

With the bang-bang control strategy in (8), a generate optimal-candidate control can be obtained. Moreover, the bang-bang control trajectory of SAVCM can be characterized by the number of switch, switching order and time segment of each switching.

Definition 7 Due to the constraints of $-\phi_{\max} \leq \phi \leq \phi_{\max}$, $-\alpha_{\max} \leq \alpha \leq \alpha_{\max}$, the maximum-accelerating-time ($T_{\max}$) can be obtained as

$$T_{\max} = \frac{\phi_{\max}}{\alpha_{\max}} \quad (9)$$

Definition 8 The following bang-bang control is called a Maxi-Minimum control (MMC)

$$\alpha_{mmc}(t, \Delta) = \begin{cases} \alpha_{\max} & [0 \ t] \\ 0 & [t \ t + \Delta] \\ -\alpha_{\max} & [t + \Delta \ 2t + \Delta] \end{cases} \quad (10)$$

Definition 9 The following orientation angle driven by $\alpha_{mmc}(T_{\max}, 0)$ is called max-turning-angle $\theta_{\max}$

$$\theta_{\max} = v \int_0^{T_{\max}} \tan(\alpha_{\max} \tau) d\tau + v \int_{T_{\max}}^{2T_{\max}} \tan(\alpha_{\max} (t_{\max} - \tau)) d\tau \quad (11)$$

Definition 10 The turning angle θ and turning length L is defined in Figure 7.

The $T_{\max}$, MMC and $\theta_{\max}$ are shown in figure 5,6. Figure 7 illustrates the difference of circle trajectory and trajectory generated by SAVCM. From the above diagrams, we can obtain the following theorem.

Theorem 1 Any turning angle $0 \leq \theta \leq \theta_{\max}$ can be reach by $\alpha_{mmc}(t, 0)$, $t \in [0 \ T_{\max}]$; Any $\theta_{\max} < \theta < \pi$ can be reach by $\alpha_{mmc}(T_{\max}, \Delta)$, $\Delta \in (0 \ \Delta_{\max})$ where $\alpha_{mmc}(T_{\max}, \Delta_{\max})$ can drive the CLR turn 2π .

Lemma 1 To a CLR which has no SAD, optimal shortest path are at most made of three pieces which are straight line segments or arcs of circle with minimal radius. See [12].

Based on the above theorem, we can generate any turning trajectory by using MMC. Because the conclusion introduced in Lemma 1, we focus our trajectory generation problem on how to generate turning trajectory which is used to replace the circle-turning trajectory.

Corollary 1 Any turning trajectory with turning angle $0 \leq \theta < \pi$ can be realized by MMC.

But we should notice that if the turning angle θ is quite near π , the turning length L is should be very long.

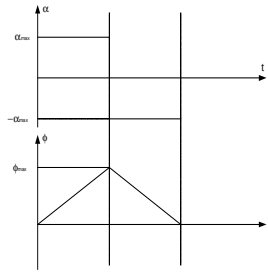


Fig. 5: (T_{Max} , MMC)

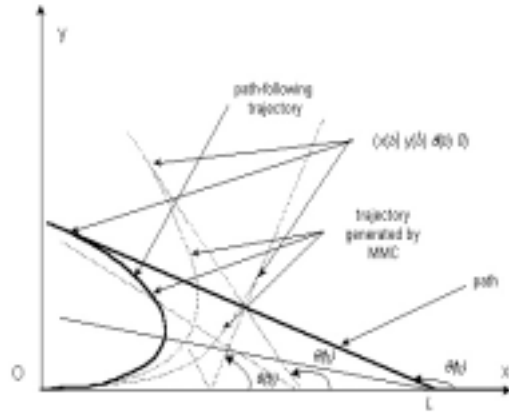


Fig.6: (Position curve of turning trajectory)

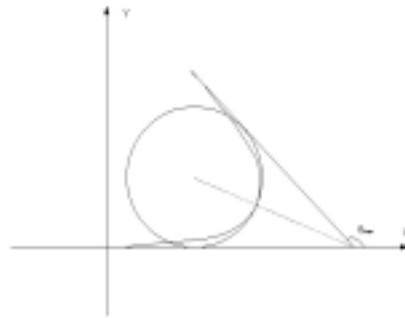


Fig.7: (Comparison of circle-trajectory and SAVCM trajectory)

Suppose the initial configuration is $(0,0,0,0)$, then the local goal configuration $(x(t_f), y(t_f), \theta(t_f), 0)$ and local trajectory can be calculated as following.

Step 1 Calculation of final time t_f

Case I: $\theta(t_f) \leq \theta_{max}$

Because

$$\theta(t_f) = v \int_0^{t_f/2} \tan(\alpha_{\max} \tau) d\tau + v \int_{t_f/2}^{t_f} \tan(\alpha_{\max} (t_f/2 - \tau)) d\tau \quad (12)$$

so t_f can be calculated by numerical approximated method[4].

Case II: $\theta(t_f) > \theta_{\max}$

Because $\theta(t_f) = \theta_{\max} + \Delta \tan \phi_{\max}$ and $t_f = 2T_{\max} + \Delta$, so Δ can be calculated easily. Then $(x(t_f), y(t_f), \theta(t_f))$ can be calculated by (1).

Step 2 Calculation of final configuration

By (12), we can get $\theta(t_f)$. Then calling (1), the values of $x(t_f), y(t_f)$ can also be calculated.

Step 3 Calculation of L (See figure 7) which will be used to determine when to start the turning-control process of $\alpha_{mmc}(T_{\max}, \Delta), \Delta \in (0, \Delta_{\max})$.

By using the above calculation results, L can be calculated as following

$$L = x(t_f) + y(t_f) \tan(\pi - \theta(t_f)) \quad (13)$$

Therefore, the final local turning trajectory characterized by L and $\alpha_{mmc}(T_{\max}, \Delta), \Delta \in (0, \Delta_{\max})$ is obtained.

The above local algorithm can be used to build a global path-following trajectory. The global trajectory generation process can be described as following:

Assume the initial position is at O where the initial configuration is $(x_0, y_0, 0, 0)$, the final point of the path P_R is F where the configuration is $(x(t_f), y(t_f), \theta(t_f), 0)$ (See figure 8). The path P_R can be described by polygon line $(O, T_1, T_2, \dots, T_{13}, F)$. The global trajectory can be generated by the following algorithm:

Step 1 For any turning (T_{i-1}, T_i, T_{i+1}) , we can use the above local trajectory generation algorithm to generate the local turning trajectory.

Step 2 For any straight path segment, the trajectory is also straight. That is the steering angle velocity equal to zero, i.e. $\alpha_i = 0$.

The figure 8 can illustrate the global trajectory generation clearly.

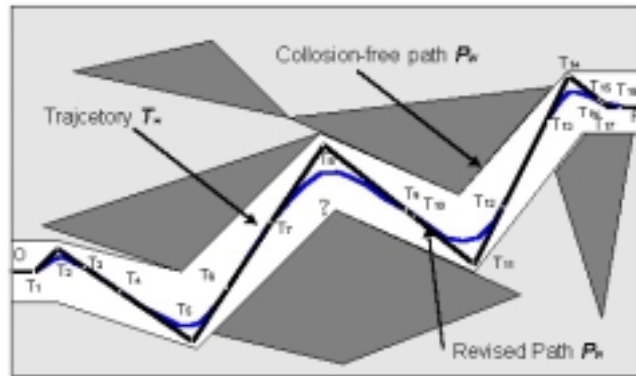


Fig.8: (Global trajectory planning)

4 Simulation Results

The CLR platform for simulation is CyCabTM. It has the following characteristics: Length - 1.9m; Width - 1.2 m; Weight - 300 kg; Power - 4 x 1 kW; electric motors; 2 wheel-drive, 1 wheel-steer; Max. speed - 36 km/h.

Simulation data:

$$\alpha_{\max} = 1 \text{ radian/sec},$$

$$\phi_{\max} = \pi/5 = 0.6283 = 36^\circ.$$

Simulation 1 In this simulation the different turning trajectories by using SACM and SAVCM are tested and compared for a CLR with significant SAD. The simulation initial data are: 1.57r/sec, i.e. $\alpha_{\max} = 1.57\text{r/sec}$ and $\phi_{\max} = \pi/5$. The simulation results are shown in Figure 3 and 4.

Simulation 2 In this simulation we test the different turning trajectory by different MMC process. Simulation results are shown in Figures 9, 10 and 11. The simulation results show that by using different MMC, i.e. the t of $\alpha_{mmc}(t,0)$ is different, we can realize different turning.

Simulation 3 In this simulation we used the algorithm presented above to realize an elliptic trajectory. The results are shown in Figure 12-14. Figure 12 illustrates the trajectory curve, Figure 13 shows the control of MMC and Figure 14 shows the steering angle of the CLR. The CLR is assumed to move with a fixed velocity of 8m/sec, the maximum of steering angle velocity is 1r/sec, the maximum of steering angle is 36° .

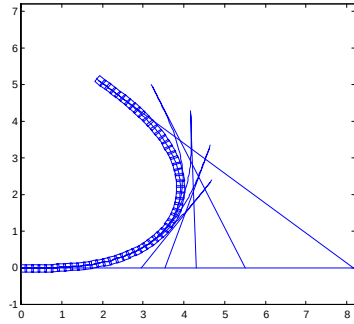


Fig.9: (Turning trajectory curve)

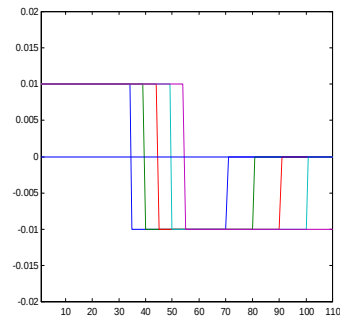


Fig.10: (MMC of CLR)

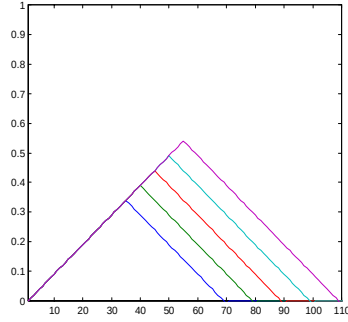


Fig.11: (Steering angles of CLR)

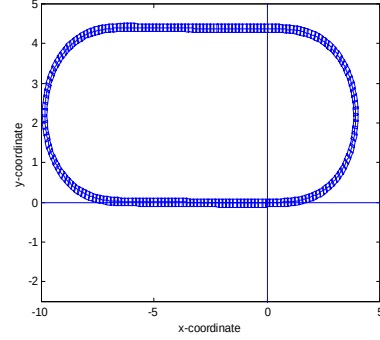


Fig.12: (Trajectory curve of CLR)

5 Conclusion

A new trajectory generation method was proposed for a car-like WMR based on bang-bang control. The effect of mv/sv ratio on the trajectory was explored. The following important points are concluded:

- (a) Mv/sv ratio is a very important factor which affect the trajectory generation for a CLR in case where the steering angle is not instantaneously realizable.
- (b) The trajectory generated by steering angle control and steering angle velocity are quite different for CLR with SAD.
- (c) The global trajectory generation based on the local trajectory generation algorithm can realize collision-free path following.

The simulation results illustrate the importance of steering angle velocity control model and proved the theoretical results in this paper.

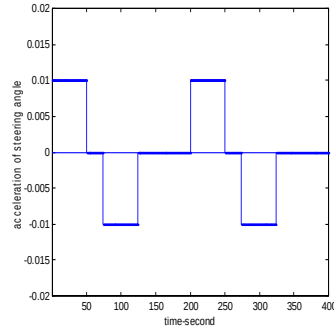


Fig. 13: (Control process of CLR)

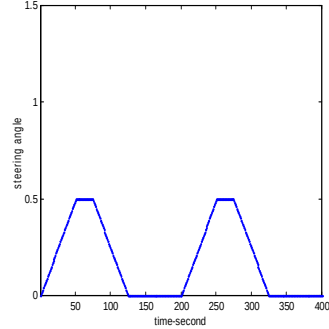


Fig.14: (Steering angle varying of CLR)

6 References

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